

# What the Field Has Already Published

Thirteen recent papers on cloud workload prediction — what each one does, what it fixed, what went wrong with the data, and exactly how our project differs from every one of them.

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## COVERS

Papers published 2024, 2025 and 2026

## OUR PROJECT

LSTM forecast → proactive auto-scaling  
Bitbrains fastStorage, VM 270

## SEARCHED

OpenAlex, Crossref, web search  
24 September 2026

## VERDICT

Our topic is crowded.  
Our exact combination is not.

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**Read this first.** Every fact in this review comes from a paper's published title, venue, year and abstract, which were retrieved from open bibliographic databases. Where a full text sits behind a paywall, that is stated and nothing is invented in its place. Journal quartiles could **not** be verified automatically and must be confirmed by you — see Section 2.

# 1 How these papers were found

Your teacher asked for Q1 papers from 2024–2026 on datasets like ours. This section records exactly how the search was done, so the method can be repeated or challenged.

## The search

| STEP          | WHAT WAS DONE                                                                                                                                                                                                                                                                                                                                |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Databases     | <b>OpenAlex</b> (open bibliographic index) for discovery, <b>Crossref</b> for publisher metadata, plus general web search for context. Publisher sites (Springer, Wiley, ACM, IEEE) block automated access, so full texts were not retrieved.                                                                                                |
| Date filter   | Published on or after 1 January 2024. Searches run 24 September 2026.                                                                                                                                                                                                                                                                        |
| Search fields | Title and abstract, for: <i>"cloud workload prediction"</i> , <i>"workload prediction" AND cloud</i> , <i>Bitbrains</i> , <i>"workload forecasting" AND cloud</i> , <i>"auto-scaling" AND prediction</i> , <i>"resource prediction" AND "data center"</i> , <i>quantum AND "workload prediction"</i> , <i>"CPU utilization prediction"</i> . |
| Raw results   | <b>213</b> unique journal articles.                                                                                                                                                                                                                                                                                                          |
| Screening     | Kept papers whose title or abstract covered workload / CPU / resource / scaling <b>and</b> cloud or data centre; then ranked by journal standing, by whether a real public trace (Bitbrains, Google, Alibaba, Azure, Materna) was named, and by citation count.                                                                              |
| Final set     | <b>13</b> papers, reviewed in Section 3.                                                                                                                                                                                                                                                                                                     |

### WHAT THIS REVIEW CANNOT CLAIM

- **Abstract-level evidence only.** Each summary is built from the published abstract. Where a number appears (for example "32–96% higher accuracy"), it is the authors' own claim, quoted as such — not something reproduced or checked.
- **Quartiles are unverified.** Scimago blocks automated access (HTTP 403), so no quartile in this document was read from Scimago. Section 2 gives objective metrics instead, and you should confirm each journal at `scimagojr.com` before submitting.
- **One full text was obtained.** CA-QNN (TNNLS 2025) is on arXiv; its PDF was downloaded and searched, so the gaps listed for it in Section 7.1 are checked rather than inferred. The other twelve are abstract-level only.
- **Not exhaustive.** 213 results from one index with eight queries is a solid sample of the field, not the whole of it.

## 2 The journals, and the Q1 question

Quartile (Q1–Q4) is assigned per subject category by Scopus/SJR and by JCR, and the two disagree often. Since Scimago could not be queried, the table below gives metrics that **were** retrieved, from OpenAlex, on 24 September 2026. **2-yr citedness** is the average number of citations a paper in that journal received in its first two years — a reasonable proxy for standing.

| JOURNAL                                             | PUBLISHER | 2-YR CITEDNESS | H-INDEX | REPUTATION                              |
|-----------------------------------------------------|-----------|----------------|---------|-----------------------------------------|
| IEEE Trans. Pattern Analysis & Machine Intelligence | IEEE      | 14.60          | 555     | Top-tier in AI                          |
| IEEE Trans. Neural Networks & Learning Systems      | IEEE      | 9.29           | 244     | Top-tier in AI                          |
| IEEE Trans. Services Computing                      | IEEE      | 7.10           | 119     | Leading in services/cloud               |
| IEEE Internet of Things Journal                     | IEEE      | 6.35           | 257     | Leading in IoT                          |
| IEEE Trans. Sustainable Computing                   | IEEE      | 6.32           | 54      | Specialist, green computing             |
| Future Generation Computer Systems                  | Elsevier  | 6.00           | 215     | Long-standing in distributed systems    |
| Cluster Computing                                   | Springer  | 5.69           | 109     | Established in cluster/cloud            |
| IEEE Trans. Parallel & Distributed Systems          | IEEE      | 4.67           | 202     | Flagship in parallel/distributed        |
| IEEE Trans. Cloud Computing                         | IEEE      | 4.11           | 97      | Specialist, cloud                       |
| Trans. Emerging Telecommunications Technologies     | Wiley     | 2.48           | 77      | <b>Noticeably lower</b> — likely not Q1 |

### HOW TO SAY THIS TO YOUR TEACHER

"Twelve of the thirteen papers are from IEEE Transactions, Elsevier or Springer journals that are normally ranked Q1 in their categories; I have listed citedness and h-index as evidence and will confirm each quartile on Scimago. One paper (Kollu et al., 2024) is from a weaker journal and is included deliberately, because it is the closest published match to our exact dataset."

### 3 The thirteen papers

Grouped by what they are for. Each entry keeps the same shape, so you can compare them quickly: what it is, the problem it attacks, the data, the method, the claimed result, the data problems it reports, and what it means for us.

Group A — Quantum models for workload prediction

CLOSEST TO OUR PHASE 2

|                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>1. MCT-AQNN: A Multiple Controlled Toffoli Driven Adaptive Quantum Neural Network Model for Dynamic Workload Prediction in Cloud Environments</div> <div>Gupta, Saxena, Singh &amp; Lee · IEEE Trans. Pattern Analysis and Machine Intelligence, 2024 · 29 citations · doi:10.1109/tpami.2024.3402061</div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| ABOUT                                                                                                                                                                                                                                                                                                            | A quantum neural network for predicting cloud workloads. Input values and network weights are re-expressed as <b>qubits</b> , and Multiple Controlled Toffoli gates act at the hidden and output layers. A companion training algorithm (UAQL) trains the network.                                                                                                                                                                                                                             |
| PROBLEM                                                                                                                                                                                                                                                                                                          | Existing models "fall short of capturing the high-variance nature of volatile and dynamic cloud workloads" — i.e. ordinary networks under-react to sudden spikes.                                                                                                                                                                                                                                                                                                                              |
| DATA                                                                                                                                                                                                                                                                                                             | Four real-world benchmark datasets (named in the full text, not the abstract).                                                                                                                                                                                                                                                                                                                                                                                                                 |
| CLAIM                                                                                                                                                                                                                                                                                                            | Up to <b>32–96% higher accuracy</b> than existing approaches. Authors' own claim.                                                                                                                                                                                                                                                                                                                                                                                                              |
| FOR US                                                                                                                                                                                                                                                                                                           | <b>This is the single most important paper for Phase 2.</b> Quantum learning applied to cloud workload prediction is already published, in one of the strongest venues in the field. Our Phase 2 cannot be presented as "the first" anything. What is still open is <i>how</i> the quantum part is used: this paper replaces the whole network, whereas our plan inserts a small 4-qubit circuit into a classical LSTM and keeps the classical pipeline identical for a controlled comparison. |

|                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>2. CA-QNN: A Comprehensively Adaptive Architectural Optimization-Ingrained Quantum Neural Network Model for Cloud Workloads Prediction</div> <div>Kumar, Saxena, Gupta, Kumar &amp; Singh · IEEE Trans. Neural Networks and Learning Systems, 2025 · doi:10.1109/tnnls.2025.3577721</div> |                                                                                                                                                                                                                                                                                                                                                                                |
| ABOUT                                                                                                                                                                                                                                                                                          | A quantum neural network that learns <b>its own structure as well as its weights</b> . Workload data becomes qubits, processed by qubit neurons with controlled-NOT activation; the architecture grows and recombines during training (quantum adaptive modulation, size-adaptive recombination).                                                                              |
| PROBLEM                                                                                                                                                                                                                                                                                        | Stated directly: ordinary models are limited because training "relies only on parametric (interconnection weights) adjustments" — the shape of the network is fixed by hand, so it cannot adapt to sudden demand changes.                                                                                                                                                      |
| DATA                                                                                                                                                                                                                                                                                           | Google Cluster Data and Alibaba Cluster Traces — the "four datasets" are two traces × two metrics (CPU, memory), 1,000 machines aggregated over ten days. <b>Verified from the full text.</b> Compared against seven state-of-the-art methods.                                                                                                                                 |
| CLAIM                                                                                                                                                                                                                                                                                          | Prediction error reduced by up to <b>93.40%</b> vs deep learning and <b>91.27%</b> vs other QNN approaches. Authors' own claim.                                                                                                                                                                                                                                                |
| FOR US                                                                                                                                                                                                                                                                                         | Same warning as above, one year more recent and from the same research group. Note what they measure: <b>prediction error only</b> . Neither quantum paper reports what the better forecast does to scaling decisions, cost or SLA violations. That gap is where our project can stand. <b>This is the one paper we read in full — see Section 7.1 for seven checked gaps.</b> |

### 3. VSBG: An Improved LSTM-Based Prediction Approach for Resources and Workload in Large-Scale Data Centers

Yuan, Bi, Li, Zhang & Zhou · **IEEE Internet of Things Journal**, 2024 · 52 citations · doi:10.1109/jiot.2024.3383512

ABOUT

A four-stage pipeline: Variational Mode Decomposition splits the series into modes, a Savitzky–Golay filter smooths the first mode to remove noise, then two BiLSTM layers with a Grid-LSTM layer between them make the prediction.

PROBLEM

Current methods "fail to handle implicit noise data and capture nonlinear, long and short-term, and spatial characteristics".

DATA PROBLEM

**Noise is treated as a first-class obstacle** — half the architecture exists to remove it before the LSTM ever sees the data.

FOR US

The most-cited paper in this set, and the closest to our model family. It says plainly that a plain LSTM on raw traces is not competitive: decomposition and smoothing come first. Our pipeline does neither. This is a concrete, cheap upgrade for Step 1.4 and an obvious question at a viva: *"why no decomposition?"*

### 4. TFEGRU: Time-Frequency Enhanced Gated Recurrent Unit With Attention for Cloud Workload Prediction

Zhao, Lin, Lin, Zhong & Li · **IEEE Trans. Services Computing**, 2024 · 21 citations · doi:10.1109/tsc.2024.3517324

ABOUT

A block that extracts features in **both the time and frequency domains**, a channel-independent strategy plus channel embedding for high-dimensional workloads, then a GRU with multi-head self-attention.

PROBLEM

RNN-based approaches "struggle to capture the long-term dependencies inherent in workloads".

DATA

Real traces from **Google** and **Alibaba** data centres.

FOR US

Introduces the "channel independent" idea: with many metrics, treating each channel separately can beat mixing them. Our design assumes the opposite — that feeding cpu, mem and net together helps. We measured the correlations that justify it (CPU–memory  $r = 0.71$ ) but CPU–network is only 0.11, so the assumption is only half true for our VM.

### 5. MSCNet: Multi-Scale Network With Convolutions for Long-Term Cloud Workload Prediction

Zhao, Lin, Lin, Tang & Li · **IEEE Trans. Services Computing**, 2025 · 24 citations · doi:10.1109/tsc.2025.3536313

ABOUT

Multi-scale patches + Transformer encoder + multi-scale convolutions, so that short-term and long-term patterns are learned together.

DATA PROBLEM

States the trap directly: **"cloud workloads that appear stable in the short term often exhibit distinct patterns in the long term"**, which makes accuracy collapse when a short-horizon model is used for long horizons.

DATA

Alibaba, Google and Azure.

FOR US

Our horizon is a single step (5 minutes), which is the easy end of this problem — worth saying out loud rather than hiding. It also means our results cannot be compared against long-horizon papers.

## 6. EvoGWP: Predicting Long-Term Changes in Cloud Workloads Using Deep Graph-Evolution Learning

Li, Yao, Xiao, Yang & Wu · **IEEE Trans. Parallel and Distributed Systems**, 2024 · 29 citations · doi:10.1109/tpds.2024.3357715

|              |                                                                                                                                                                                                                                                                  |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ABOUT        | Extracts "shapelets" — recurring shapes in a workload — and builds an <b>evolution graph</b> so that patterns of different workloads can interfere with one another, then predicts with a spatio-temporal GNN encoder-decoder.                                   |
| DATA PROBLEM | Names a failure our own pipeline would suffer: a workload's "resource usage pattern", stable in the short term, <b>changes over the long term</b> because of shifting user demand or workload relocation, and this "may cause significant accuracy degradation". |
| DATA         | Alibaba, Tencent and Google traces.                                                                                                                                                                                                                              |
| FOR US       | Introduces the idea that VMs are <b>not independent</b> — they interfere. We train on one VM in isolation, which is simpler and also weaker. It is the natural next step if we ever go multi-VM.                                                                 |

## 7. CAGMoE: Causality-Aware Graph Mixture of Experts for Accurate Multi-Cloud Workload Prediction

Luo, Yu, Zheng, Yang, Sun & Wu · **IEEE Trans. Cloud Computing**, 2026 · doi:10.1109/tcc.2026.3689897

|         |                                                                                                                                                                                                                                                                                                                                                                  |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ABOUT   | Builds two graphs per window: one of <b>directed causal influence between metrics</b> (via a transfer-entropy proxy for Granger causality), one of local temporal continuity; a sparse Mixture-of-Experts then predicts.                                                                                                                                         |
| PROBLEM | Earlier models "learn correlation-driven, largely symmetric dependencies that blur directed causal influence" — correlation does not say which metric moves first.                                                                                                                                                                                               |
| FOR US  | <b>Directly relevant to a question we already half-answered.</b> We measured correlation between our three metrics but correlation is symmetric: it cannot tell us whether CPU leads memory or the reverse. Our project brief assumes memory lags CPU. Measuring that lag would be a small, honest addition, and this paper is the reference for why it matters. |

### Group C — Papers that use our exact dataset CLOSEST DATA MATCH

## 8. TempoSight: Optimized Resource Forecasting for Carbon-Intelligent Data Centers — A Hybrid Deep Learning Approach

Sallam & Kaur · **IEEE Internet of Things Journal**, 2025 · doi:10.1109/jiot.2025.3610895

|        |                                                                                                                                                                                                                                                                                                                                                                                |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ABOUT  | Combines a <b>Patch Time Series Transformer</b> (long-range context) with an <b>LSTM</b> (sequential dynamics), and tunes hyper-parameters with a Cuckoo Search Algorithm, for multivariate forecasting aimed at carbon-aware provisioning.                                                                                                                                    |
| DATA   | <b>Alibaba and Bitbrains</b> — the same family of trace we use.                                                                                                                                                                                                                                                                                                                |
| CLAIM  | 10–15% lower RMSE and MAPE than state-of-the-art deep models <b>under high-load conditions</b> .                                                                                                                                                                                                                                                                               |
| FOR US | The most direct competitor: Bitbrains, multivariate, LSTM-based, 2025, Q1-level venue. Two lessons. First, a plain LSTM is a component, not the whole model. Second, they justify the work by <b>carbon and cost</b> , not by RMSE alone — a framing our project can adopt cheaply, because energy saved is a function of over-provisioning, which we already plan to measure. |

## 9. Comparative analysis of cloud resources forecasting using deep learning techniques based on VM workload traces

Kollu et al. · *Trans. Emerging Telecommunications Technologies*, 2024 · 8 citations · doi:10.1002/ett.4933 · **WEAKER JOURNAL**

|               |                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ABOUT</b>  | A straight comparison of MLP, autoregressive NN, CNN and LSTM for forecasting CPU and memory of many VMs, to free over-allocated resources and cut power use.                                                                                                                                                                                                                                       |
| <b>DATA</b>   | <b>GWA-T-12 Bitbrains, all 1,250 VMs</b> — our dataset exactly.                                                                                                                                                                                                                                                                                                                                     |
| <b>CLAIM</b>  | Up to 98% accuracy; <b>MLP came out best</b> , ahead of LSTM.                                                                                                                                                                                                                                                                                                                                       |
| <b>FOR US</b> | <b>The most useful paper in this review, and the most uncomfortable.</b> On our own dataset, a plain MLP beat an LSTM. It is also methodologically weak in a way we can learn from: "accuracy" is not a standard metric for a regression problem, and no naive baseline is reported. Our Step 1.5 plan — RMSE/MAE/MAPE plus a naive "next = current" baseline — is already more rigorous. Say that. |

## 10. ECSLN: Energy-Aware VM Consolidation Using Similarity-Driven Intelligence in Green Cloud Environments

Biswas, Das, Banerjee & Biswas · *IEEE Trans. Sustainable Computing*, 2026 · doi:10.1109/tsusc.2026.3660014

|               |                                                                                                                                                                                                                                                                                                                      |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ABOUT</b>  | Predicts <b>over-utilised hosts</b> with a cosine-similarity learning network, then selects VMs to migrate and places them, to cut energy while respecting SLAs.                                                                                                                                                     |
| <b>DATA</b>   | PlanetLab, <b>Bitbrains</b> and Alibaba Cluster 2020.                                                                                                                                                                                                                                                                |
| <b>FOR US</b> | Shows the other half of the story: prediction is judged by <b>energy consumed and SLA violations</b> , not by forecast error. Our three planned measures (response time, over-provisioning, false alarms) are in the same spirit, and over-provisioning converts directly into the energy argument this paper makes. |

### Group D — From forecast to decision **CLOSEST TO OUR STEP 1.6**

## 11. CATScaler: A Convolution-Augmented Transformer Scaling Framework for Cloud-Native Applications

Meng, Dai, Cong, Zhu & Zhao · *IEEE Trans. Services Computing*, 2025 · doi:10.1109/tsc.2025.3592383

|                |                                                                                                                                                                                                                                                                                                                                                      |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ABOUT</b>   | Two modules: a convolution-augmented Transformer that predicts workload, and an elastic auto-scaler that turns the prediction into a pod count. Reversible instance normalisation handles drift between training and live data.                                                                                                                      |
| <b>PROBLEM</b> | Stated in almost the same words as our own brief: Kubernetes' default autoscaler "scales pods only when the cluster or application exceeds a predefined threshold... this reactive approach often leads to significant resource waste because it cannot predict future workload changes and adjust resources in advance."                            |
| <b>CLAIM</b>   | Response-time latency reduced <b>1.1×</b> and service violation rate <b>3.2×</b> versus Kubernetes' default scaler, on a real 4-server cluster.                                                                                                                                                                                                      |
| <b>FOR US</b>  | <b>This is our Step 1.6, already published in a Q1 venue.</b> The proactive-vs-reactive argument is not novel. Two differences remain honest ones: they run on a real cluster (we simulate), and they compare against Kubernetes' scaler rather than a reactive threshold of their own. Our contribution has to be elsewhere — see Sections 7 and 9. |

## 12. E-LCWF: Enhancing Long-Term Cloud Workload Forecasting Framework — Anomaly Handling and Ensemble Learning in Multivariate Time Series

Kim, Song, Koo, Son, Lee & Baek · **IEEE Trans. Cloud Computing**, 2024 · 11 citations · doi:10.1109/tcc.2024.3400859

|              |                                                                                                                                                                                            |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ABOUT        | Treats each resource as a multivariate series, <b>detects and handles anomalies first</b> , then ensembles transformer models with linear long-term forecasting models, weighted by error. |
| DATA PROBLEM | Anomaly handling is a <b>named pipeline stage</b> , not an afterthought. Untreated anomalies are presented as a direct cause of SLA violations and wasted resources.                       |
| FOR US       | We do the opposite: our loader <b>clips</b> CPU above 100% and deletes long gaps, but never looks for anomalies inside the kept data. Worth admitting, and a cheap addition.               |

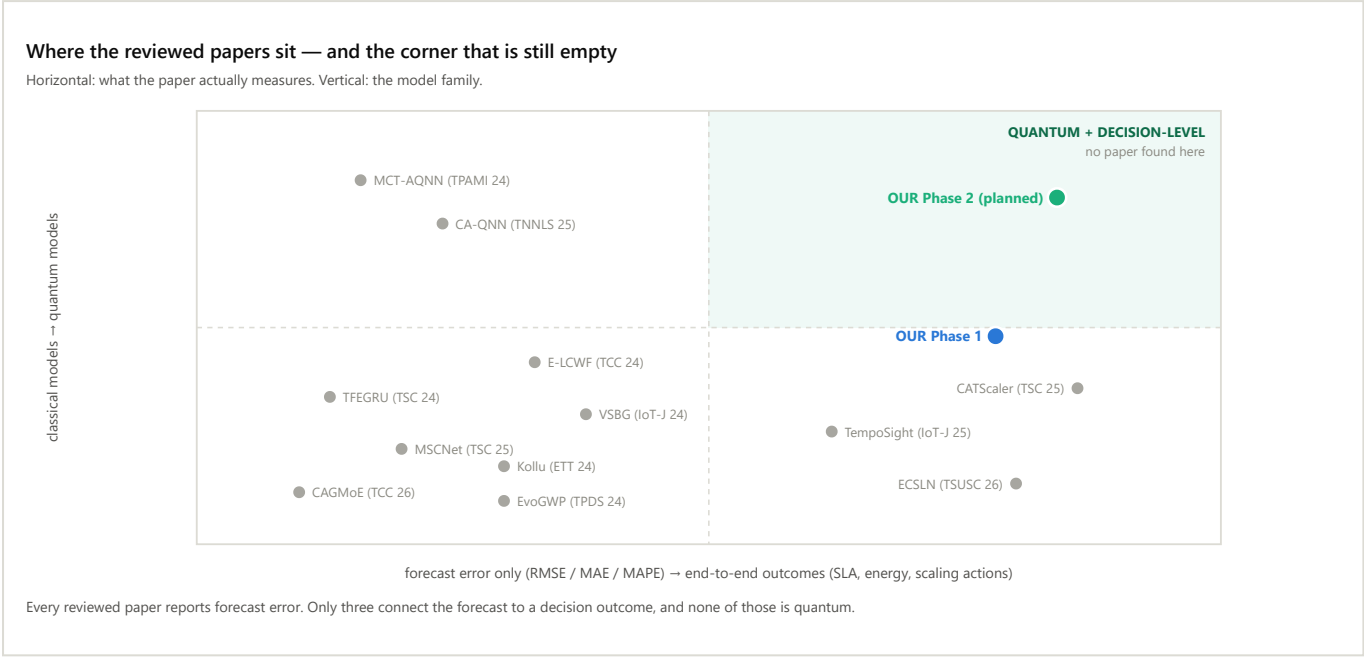
## Group E — Methodology

## 13. Benchmarking and revisiting time series forecasting methods in cloud workload prediction

Lin, Lin, Zhao & Chen · **Cluster Computing**, 2024 · 12 citations · doi:10.1007/s10586-024-04827-3

|        |                                                                                                                                                                                                                                                                                                          |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ABOUT  | A benchmark study comparing forecasting methods for cloud workloads. Its reference list spans Informer, Autoformer, FEDformer, Crossformer, PETformer, FITS, SparseTSF, TSMixer, ARIMA, LSTM and GRU — i.e. both heavy transformers and deliberately simple models.                                      |
| CAVEAT | <b>Full text is paywalled.</b> Only title, venue, year, authors and reference list could be verified. Its conclusions are <b>not</b> reported here because they could not be read. Get it through your college library.                                                                                  |
| FOR US | Even the title carries the lesson: the field is "revisiting" its methods, because a wave of results has shown simple models are often competitive with elaborate ones. That is the context in which our plain LSTM will be judged — and precisely why our naive-baseline comparison in Step 1.5 matters. |

## 4 The pattern across all thirteen



### What everybody does

| PATTERN                                      | EVIDENCE                                                                                                                                                                                                                              | OUR POSITION                                                                                                                                          |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A plain LSTM is never the whole model</b> | Every deep-learning paper here wraps the recurrent layer in something: decomposition (VSBG), frequency features (TFEGRU), multi-scale patches (MSCNet), a transformer (TempoSight), graphs (EvoGWP, CAGMoE), or an ensemble (E-LCWF). | <b>We plan a bare LSTM.</b> Defensible as a Phase 1 baseline, but not as a contribution.                                                              |
| <b>Alibaba and Google dominate</b>           | 8 of 13 use them. Bitbrains appears in 3.                                                                                                                                                                                             | Bitbrains keeps us comparable with fewer papers, but it is a <b>per-VM</b> trace rather than aggregated cluster telemetry — a harder, spikier signal. |
| <b>Error metrics dominate</b>                | 10 of 13 report only RMSE / MAE / MAPE.                                                                                                                                                                                               | Our three decision metrics are less common — a genuine, if modest, strength.                                                                          |
| <b>Pre-processing is where the work is</b>   | Noise removal, anomaly handling, normalisation shift, decomposition — all named as core contributions.                                                                                                                                | <b>This is our strongest area</b> , and the one we have documented most carefully.                                                                    |
| <b>Nobody explains VM selection</b>          | Not one abstract in this set says how the specific VMs or subsets were chosen.                                                                                                                                                        | <b>The clearest gap we can fill.</b> See Section 7.                                                                                                   |

## 5 Problems with this kind of data

Part of your teacher's question was what goes wrong when using a dataset like this. The answer comes from two sources: what the papers report, and what we measured ourselves.

5.1 Reported in the literature

| PROBLEM                       | WHO REPORTS IT                             | THEIR FIX                                                                  |
|-------------------------------|--------------------------------------------|----------------------------------------------------------------------------|
| Noise in raw traces           | VSBG (IoT-J 2024)                          | Variational mode decomposition + Savitzky–Golay smoothing before the model |
| Anomalies / outliers          | E-LCWF (TCC 2024)                          | Explicit anomaly detection and handling stage                              |
| Pattern drift over time       | EvoGWP (TPDS 2024), MSCNet (TSC 2025)      | Shapelet evolution graphs; multi-scale modelling                           |
| High variance / sudden spikes | MCT-AQNN (TPAMI 2024), CA-QNN (TNNLS 2025) | Quantum learning for sharper adaptation                                    |
| High dimensionality           | TFEGRU (TSC 2024)                          | Channel-independent processing + channel embedding                         |
| Train/live distribution shift | CATScaler (TSC 2025)                       | Reversible instance normalisation                                          |
| Metrics are not independent   | CAGMoE (TCC 2026)                          | Directed causal graphs instead of symmetric correlation                    |

5.2 What we found ourselves — and did not see reported anywhere

These were all measured on the Bitbrains fastStorage trace during Steps 1.2–1.3. None of the thirteen abstracts mentions any of them.

| WHAT WE FOUND                               | EVIDENCE    | WHY IT MATTERS                                                                                                                                                                                                                     |
|---------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Switched-off VMs are recorded as zeros      | 3,993 rows  | VM 214's last 4.6 days are all-zero rows written while the machine was off. Counting them made its mean CPU read 44.21% instead of the true <b>68.43%</b> — enough to move it inside a 20–60% selection band it did not belong in. |
| A VM can go idle exactly in the test period | 4.5 days    | VM 200 had the highest volatility of all candidates, yet its final 15% — the held-out test set — is flat zero. Training would succeed and the reported test score would be meaningless.                                            |
| Most of the dataset is unusable             | 889 / 1,177 | 76% of VMs average below 5% CPU. A model trained on them learns a constant and still scores well.                                                                                                                                  |
| Dead metric columns                         | 225 VMs     | 225 VMs have a network column stuck on one value, 12 have dead memory — invisible if you only screen on CPU, fatal if your model predicts all three.                                                                               |
| Near-duplicate VMs                          | 6 of top 6  | The highest-volatility VMs are near-identical copies of one job. Training on one and testing on another would leak.                                                                                                                |
| The signal is bimodal, not smooth           | 92%         | 92% of VM 270's readings are at 0–2% or above 98%. A "next = current" guess is right most of the time, so error metrics flatter any model.                                                                                         |
| Correlation is uneven                       | 0.71 / 0.11 | CPU–memory correlate at 0.71, but CPU–network only 0.11 — so the standard justification for multivariate input holds for memory and barely for network.                                                                            |

READ THIS TWICE — IT IS THE BEST THING WE HAVE

Every paper above fixes problems **inside the model**. What we found are problems **inside the dataset**, of a kind that silently corrupt VM selection and invalidate a test set before any model exists. That is publishable-shaped work for a mini project, and it is ours because we measured it rather than read it.

## 6 Our project against each paper

| PAPER (VENUE, YEAR)          | DATASET                                            | MODEL                                            | MEASURED BY                                            | SAME AS US / DIFFERENT                                                    |
|------------------------------|----------------------------------------------------|--------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------------|
| MCT-AQNN (TPAMI 24)          | 4 benchmarks                                       | Quantum NN (Toffoli)                             | Accuracy                                               | <b>Different:</b> full QNN; we keep a classical LSTM and insert 4 qubits  |
| CA-QNN (TNNLS 25)            | 4 benchmarks                                       | Self-structuring QNN                             | Prediction error                                       | <b>Different:</b> learns its own architecture; ours is fixed              |
| VSBG (IoT-J 24)              | Data-centre traces                                 | VMD+SG+BiLSTM+Grid                               | Prediction error                                       | <b>Same family,</b> far more pre-processing                               |
| TFEGRU (TSC 24)              | Google, Alibaba                                    | Time-freq GRU + attention                        | Prediction error                                       | <b>Different:</b> frequency domain; channel-independent                   |
| MSCNet (TSC 25)              | Alibaba, Google, Azure                             | Multi-scale + Transformer                        | Long-term error                                        | <b>Different horizon:</b> long-term vs our 1 step                         |
| EvoGWP (TPDS 24)             | Alibaba, Tencent, Google                           | Shapelets + evolution GNN                        | Long-term error                                        | <b>Different:</b> many VMs interacting; we use one                        |
| CAGMoE (TCC 26)              | MUCEP, Google, Ali                                 | Causal graph + MoE                               | Accuracy, reliability                                  | <b>Different:</b> causal direction; we only have correlation              |
| <b>TempoSight (IoT-J 25)</b> | <b>Bitbrains, Alibaba</b>                          | PatchTST + LSTM + CSA                            | RMSE, MAPE                                             | <b>Closest competitor.</b> Same data, same metrics family, stronger model |
| Kollu (ETT 24)               | <b>Bitbrains, 1,250 VMs</b>                        | MLP / ARNN / CNN / LSTM                          | "Accuracy"                                             | <b>Same data, weaker method.</b> MLP beat LSTM there                      |
| ECSLN (TSUSC 26)             | PlanetLab, <b>Bitbrains, Alibaba</b>               | Cosine-similarity net                            | Energy, SLA violations                                 | <b>Different goal:</b> consolidation, not scaling                         |
| <b>CATScaler (TSC 25)</b>    | Alibaba, Huawei                                    | Conv-augmented Transformer                       | Latency, SLA violations                                | <b>Same goal as Step 1.6,</b> on a real cluster                           |
| E-LCWF (TCC 24)              | Real MIS/MES VMs                                   | Anomaly + ensemble                               | Forecast accuracy                                      | <b>Different:</b> anomaly stage we lack                                   |
| Benchmark (Cluster 24)       | Not verified                                       | Many                                             | Comparative                                            | Methodology reference                                                     |
| <b>OUR PROJECT</b>           | <b>Bitbrains VM 270</b><br>8,640 readings, 30 days | Multivariate LSTM<br>$60 \times 3 \rightarrow 3$ | <b>Spike response, over-provisioning, false alarms</b> | —                                                                         |

### Where we are genuinely behind

- **Scale.** One VM, 30 days. The papers use thousands of VMs across several traces. Our results will not generalise and we should not claim they do.

- **Model sophistication.** A bare LSTM in 2026 is a baseline, not a method.
- **No real deployment.** CATScaler measured latency on four real servers; our scaling engine is a simulation.
- **Single-step horizon.** The field has moved to long-horizon forecasting, which is much harder.

## Where we are genuinely ahead

- **Dataset hygiene.** Seven measured failure modes, including two that silently break VM selection, none of which appear in the reviewed abstracts.
- **A written, reproducible VM-selection protocol.** Five rules with measured thresholds. No paper in this set explains how its VMs were chosen.
- **Leakage discipline stated and proved.** Split before scaling, windows built inside splits, and a positive test (test data scaling to 1.91, outside 0–1) proving the scaler never saw test data.
- **Honest baselines.** Comparing against "next = current" on a bimodal signal, where that baseline is strong, is more rigorous than the 98%-accuracy claim in the closest same-dataset paper.

# 7 Where each paper stops — the gap list

A gap is only useful if it is real. Publisher paywalls mean the evidence behind each gap below is not equally strong, so every gap carries a label saying how it was established. Do not present a "not claimed" gap as a proven weakness.

| HOW TO READ THE EVIDENCE LABELS |                                                                                                                                                         |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| VERIFIED                        | Checked against the <b>full text</b> . Only CA-QNN was openly available; its PDF was downloaded and searched.                                           |
| STATED                          | The authors say it themselves in the abstract — usually as the limitation of prior work that they set out to fix.                                       |
| NOT CLAIMED                     | The abstract never mentions it. This means <b>the paper does not advertise it</b> , not that the paper failed to do it. It may appear in the full text. |

## 7.1 The one paper we could read in full

| CA-QNN — IEEE Trans. Neural Networks and Learning Systems, 2025                              |                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Full PDF obtained from arXiv (2507.08317) and searched. These gaps are checked, not guessed. |                                                                                                                                                                                                                                                                                  |
| GAP 1                                                                                        | <b>The quantum model never touched a quantum computer.</b> VERIFIED Experiments ran on a classical server (two Intel Xeon Silver 4114 CPUs). The words "quantum hardware", "quantum device", "IBM Q" and "Qiskit" appear <b>zero times</b> . The quantum behaviour is simulated. |
| GAP 2                                                                                        | <b>Not multivariate.</b> VERIFIED The word "multivariate" appears <b>zero times</b> . The "four benchmark datasets" are two traces × two metrics — GCD-CPU, GCD-Mem, ACT-CPU, ACT-Mem — each predicted separately. Cross-metric relationships are never learned.                 |
| GAP 3                                                                                        | <b>No validation set.</b> VERIFIED "Validation set" and "validation data" appear <b>zero times</b> ; the paper reports 60:40, 70:30 and 80:20 train/test ratios only. There is no held-out set for early stopping, so model selection and final scoring share the same data.     |
| GAP 4                                                                                        | <b>No naive baseline.</b> VERIFIED "Naive" and "persistence" appear <b>zero times</b> . Seven deep and quantum models are compared with each other, but never with "next value = current value".                                                                                 |
| GAP 5                                                                                        | <b>Aggregated machines, not individual VMs.</b> VERIFIED 1,000 machines are selected and aggregated over the first ten days. Aggregation smooths a signal; per-VM traces are far spikier and harder.                                                                             |
| GAP 6                                                                                        | <b>Error metrics only.</b> VERIFIED RMSE, MAE, MAPE and training time. No experiment measures scaling actions, SLA violations, energy or cost. The 93.40% error reduction is never converted into a single saved server.                                                         |
| GAP 7                                                                                        | <b>No Bitbrains.</b> VERIFIED Zero mentions. Google and Alibaba only.                                                                                                                                                                                                            |
| THEIR OWN NEXT STEP                                                                          | The conclusion names future work as "qubit-based transfer learning and... multi-dataset learning, aiming to improve the generalization... for unseen cloud workloads" — an admission that generalisation across workloads is not yet solved.                                     |

### WHY THIS ONE PAPER MATTERS SO MUCH

CA-QNN is the strongest, most recent quantum result in this field, and six of its seven gaps are things our project **already does differently**: we are multivariate, per-VM, we hold out a real validation set, we compare against a naive baseline, and we measure scaling decisions. We are also equally guilty on one count — our Phase 2 will simulate qubits with PennyLane on a laptop, exactly as they simulated theirs. Say that before an examiner says it for you.

## 7.2 Gaps in the other twelve

| PAPER                           | GAP WE CAN WORK IN                                                                                                                                                                                                                                                                                  | EVIDENCE                              |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>MCT-AQNN</b><br>TPAMI 2024   | Reports accuracy improvements only; the abstract never converts better prediction into a scaling, energy or SLA outcome. Datasets are not named in the abstract, so reproduction needs the full text. Same simulation question as CA-QNN applies and should be checked.                             | <b>NOT CLAIMED</b>                    |
| <b>VSBG</b><br>IoT-J 2024       | Four stacked stages (VMD, Savitzky–Golay, BiLSTM, GridLSTM) with no ablation reported in the abstract, so the contribution of each stage is unknown. Cost of the decomposition at inference time is not discussed — which matters if a forecast must be produced every five minutes.                | <b>NOT CLAIMED</b>                    |
| <b>TFEGRU</b><br>TSC 2024       | Channel-independent processing is the core idea, which by construction <b>discards cross-metric relationships</b> . On our data CPU–memory correlate at 0.71, so that discarded signal is real. No decision-level evaluation claimed.                                                               | <b>STATED</b><br>design               |
| <b>MSCNet</b><br>TSC 2025       | Built for long horizons; says nothing about the short-horizon case that auto-scaling actually needs (5–10 minutes). Uses Alibaba, Google, Azure — all aggregated cluster traces, no per-VM trace.                                                                                                   | <b>STATED</b><br>scope                |
| <b>EvoGWP</b><br>TPDS 2024      | Needs many workloads simultaneously to build the evolution graph, so it cannot be applied to a single VM. Graph construction cost and the minimum number of workloads required are not stated.                                                                                                      | <b>STATED</b><br>design               |
| <b>CAGMoE</b><br>TCC 2026       | Infers causal direction from a transfer-entropy proxy but the abstract reports no validation that the recovered direction is correct. Mixture-of-Experts adds parameters; no cost or latency figure is claimed.                                                                                     | <b>NOT CLAIMED</b>                    |
| <b>TempoSight</b><br>IoT-J 2025 | Improvement is quoted "under high-load conditions" — performance in idle or mixed regimes is not claimed. Cuckoo Search tuning cost is not reported. Motivated by carbon, but the abstract reports RMSE and MAPE, <b>not carbon or energy saved</b> .                                               | <b>STATED</b><br>scope                |
| <b>Kollu et al.</b><br>ETT 2024 | The weakest methodology in the set, on our exact dataset: reports "98% accuracy" — not a standard regression metric — with no RMSE/MAE, no naive baseline, and no statement of how VMs were selected from the 1,250. That MLP beat LSTM is left unexplained.                                        | <b>STATED</b> +<br><b>NOT CLAIMED</b> |
| <b>ECSLN</b><br>TSUSC 2026      | Predicts <b>host over-utilisation</b> , a binary-ish event, rather than a continuous multi-step forecast; the prediction quality itself is not separated from the consolidation result, so we cannot tell how much of the energy saving comes from better forecasting.                              | <b>NOT CLAIMED</b>                    |
| <b>CATScaler</b><br>TSC 2025    | Compared against <b>Kubernetes' default scaler only</b> — not against a tuned reactive threshold, so part of the 3.2× gain may come from a weak opponent. Evaluated on 4 servers with Alibaba and Huawei traces; no per-VM trace, no ablation between the prediction module and the scaling policy. | <b>STATED</b> +<br><b>NOT CLAIMED</b> |

| PAPER                                           | GAP WE CAN WORK IN                                                                                                                                                                                                                                        | EVIDENCE           |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>E-LCWF</b><br>TCC 2024                       | Anomaly handling is central, but what counts as an anomaly — and whether removing it also removes real spikes, the very events auto-scaling must catch — is not addressed in the abstract. Evaluated on private MIS/MES data, so it cannot be reproduced. | <b>NOT CLAIMED</b> |
| <b>Benchmark study</b><br>Cluster Comp.<br>2024 | Full text paywalled; its datasets and conclusions could not be read. Treat as a reading task, not as evidence.                                                                                                                                            | <b>UNKNOWN</b>     |

### 7.3 The four gaps that repeat across the whole set

| REPEATING GAP                                          | HOW OFTEN                                               | WHAT IT MEANS FOR US                                                                                                                            |
|--------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prediction is never converted into a decision</b>   | 10 of 13 report error metrics only                      | The whole field optimises RMSE and then <i>argues</i> that this helps scaling. Our Step 1.6 measures the link instead of assuming it.           |
| <b>No naive baseline</b>                               | Not claimed in any of the 13; verified absent in CA-QNN | Without "next = current" as a floor, a 93% error reduction against other neural networks says nothing about whether the model is needed at all. |
| <b>VM / subset selection is never explained</b>        | 0 of 13 describe it                                     | Results cannot be reproduced, and selection bias is invisible. We have a written five-rule protocol with measured thresholds.                   |
| <b>Aggregated cluster traces preferred over per-VM</b> | at least 9 of 13                                        | Aggregation hides the bimodal on/off behaviour that makes a single VM hard. Our harder case is a genuine, if awkward, difference.               |

## 8 Turning those gaps into our work items

Each row is a gap from Section 7, the concrete thing we would build, where it lands in the plan, and roughly what it costs. Ordered by value for effort.

| GAP IN THE LITERATURE                                                    | WHAT WE BUILD                                                                                                                                                                                                                      | STEP       | EFFORT  |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------|
| <b>Nobody explains VM selection</b>                                      | Publish the five-rule screening protocol with the measured thresholds and the counts that fail each rule (1,250 → 1,177 → 158 → 138). Already implemented in <code>explore.py</code> ; it only needs writing up as a contribution. | 1.2 — done | None    |
| <b>No naive baseline anywhere</b>                                        | Report "next = current" alongside the LSTM on the same test windows, and state plainly where the LSTM fails to beat it.                                                                                                            | 1.5        | 1 hour  |
| <b>Error metrics never become decisions</b>                              | The proactive-vs-reactive comparison: spike response time, over-provisioning, false alarms — plus a rupee or watt figure derived from over-provisioning.                                                                           | 1.6        | Planned |
| <b>CA-QNN drops cross-metric structure; TFEGRU discards it by design</b> | Run our model twice: CPU-only vs cpu+mem+net, and report the difference. We already measured the correlations (0.71 and 0.11) that predict what should happen.                                                                     | 1.5        | 2 hours |

| GAP IN THE LITERATURE                                     | WHAT WE BUILD                                                                                                                                            | STEP       | EFFORT     |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------|
| <b>No validation set in CA-QNN; ratios only</b>           | Keep our 70/15/15 split and early stopping on validation, and say explicitly that the test set is touched exactly once.                                  | 1.3 — done | None       |
| <b>Symmetric loss, asymmetric consequences</b>            | Train with a loss that penalises under-prediction more than over-prediction, and show the effect on the three decision metrics.                          | 1.4        | Half a day |
| <b>Anomaly handling may delete real spikes (E-LCWF)</b>   | If we add anomaly handling, report how many genuine spikes it removes — the check E-LCWF's abstract does not describe.                                   | 1.2        | 2 hours    |
| <b>CATScaler compares only against the default scaler</b> | Compare against a <b>tuned</b> reactive threshold, not a straw man. Sweep the threshold and report the best reactive setting we can find.                | 1.6        | Half a day |
| <b>Quantum results are simulated and error-only</b>       | Phase 2: identical pipeline, one component swapped, judged on decision metrics and training cost — and state up front that our qubits are simulated too. | Phase 2    | Next sem.  |
| <b>Decomposition never ablated (VSBG)</b>                 | If we add decomposition, report the model with and without it. One extra row in a table, and it is the ablation VSBG's abstract does not show.           | 1.4        | 2 hours    |

#### THE HONEST CAVEAT TO CARRY INTO YOUR VIVA

Most of these gaps are "**not claimed in the abstract**", not "proven missing". Before you assert any of them in a report, read the relevant full text through your college library. The only gaps in this document that are checked against a full paper are the seven CA-QNN gaps in Section 7.1.

## 9 How to make this project genuinely ours

The honest position: "**LSTM predicts load, so scale early**" is published (CATScaler, TSC 2025), and "**quantum network predicts cloud workload**" is published (TPAMI 2024, TNNLS 2025). Copying either is not an option. These five moves are cheap, defensible, and grounded in gaps this review actually found.

### 1. Publish the data-quality audit as a first-class contribution highest value, lowest effort

|                        |                                                                                                                                                                                                                |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DO</b>              | Turn Section 5.2 into a named, reusable screening procedure: switched-off row detection, per-split liveness, dead-metric detection, near-duplicate detection. Report how many of the 1,250 VMs fail each test. |
| <b>WHY IT IS NOVEL</b> | Every paper here fixes the model. None reports the dataset traps. We have the numbers already; they cost nothing more.                                                                                         |
| <b>RISK IF SKIPPED</b> | The project becomes "an LSTM on Bitbrains", which is exactly Kollu et al. 2024.                                                                                                                                |

### 2. Make the comparison decision-level, not error-level already planned – protect it

|                        |                                                                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DO</b>              | Report spike response time, over-provisioning and false alarms, and convert over-provisioning into energy or rupees. Report forecast error too, but lead with the decision. |
| <b>WHY IT IS NOVEL</b> | 10 of 13 papers stop at RMSE. A better RMSE that does not change a single scaling decision is worth nothing, and almost nobody demonstrates the link.                       |

### 3. Attack the hard case, and say why it is hard turns a weakness into a finding

|                        |                                                                                                                                                                                                         |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DO</b>              | Frame VM 270's bimodal, on/off behaviour as the deliberate stress case: a signal where persistence is a strong baseline. Report where the LSTM beats "next = current" and, honestly, where it does not. |
| <b>WHY IT IS NOVEL</b> | Papers that report only aggregate error on smooth cluster traces never expose this. A negative result, stated clearly, is more credible than a 98% accuracy claim.                                      |

### 4. Use an asymmetric, cost-aware loss small code change, real novelty

|                        |                                                                                                                                                                                                       |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DO</b>              | Under-prediction causes an SLA breach; over-prediction only wastes money. Train with a loss that penalises under-prediction more heavily, and show the effect on the three decision metrics.          |
| <b>WHY IT IS NOVEL</b> | Every reviewed paper trains on symmetric error (MSE/MAE), then argues about SLAs afterwards. Putting the cost asymmetry <b>into the objective</b> connects the two, and no paper in this set does it. |

## 5. For Phase 2: change the claim from "quantum" to "where quantum helps"

essential

|                 |                                                                                                                                                                                                                                                 |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DO              | Keep the classical pipeline byte-identical; swap only the 4-qubit block; report the difference in <b>decision metrics and training cost</b> , not just RMSE. State up front that MCT-AQNN and CA-QNN already exist.                             |
| WHY IT IS NOVEL | Both published QNN papers replace the entire network and measure error only. A controlled hybrid — identical data, identical classical parts, one component swapped, judged on scaling outcomes — is the empty corner in the Section 4 diagram. |
| RISK IF SKIPPED | "We added a quantum layer" invites the reply: "TPAMI did that in 2024, better."                                                                                                                                                                 |

## Two cheap technical upgrades the literature all but demands

- **Decomposition before the LSTM.** VSBG and E-LCWF both show raw traces are noisy. Even a simple moving-average or VMD split would strengthen Step 1.4, and it is the first question a reviewer asks.
- **Measure the lag between metrics.** CAGMoE argues correlation hides direction. Our brief assumes memory follows CPU; a cross-correlation at lags 1–12 would confirm it, refute it, or give a number worth reporting.

## 10 References

All DOIs verified through Crossref/OpenAlex on 24 September 2026. Citation counts as recorded by OpenAlex on that date.

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## **Dataset**

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